Modeling Glycohemoglobin Variation with Hierarchical Bayesian Methods

Project Report

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Contents

1 Introduction

Effective prevention and early treatment of diabetes rely heavily on identifying reliable early-onset markers before the disease fully develops. Glycohemoglobin (GH), a widely recognized clinical indicator of long-term blood glucose regulation, plays a central role in this effort, yet its determinants can vary substantially across social and demographic groups. In particular, unequal access to healthcare, often reflected through differences in income, as well as genetic differences between racial groups, may influence not only GH levels but also the factors that best predict them. Understanding whether prediction patterns differ across population subsets therefore holds practical importance: if the drivers of elevated GH vary by group, preventive strategies and early interventions must be tailored accordingly.

To investigate these questions, we use Bayesian modeling with brms to examine how glycohemoglobin can be predicted by factors such as age and waist length and how these relationships vary between different income and racial groups. We begin with a simple intercept-only model as a baseline, followed by a non-hierarchical model incorporating key physiological predictors. We then extend this framework with three hierarchical models that allow predictor effects to vary between income groups, racial categories, and the combination of both. Comparing these models enables us to assess whether the strength or direction of predictive factors depends on subgroup membership, and ultimately whether improved stratification can enhance early detection of individuals at risk for diabetes.

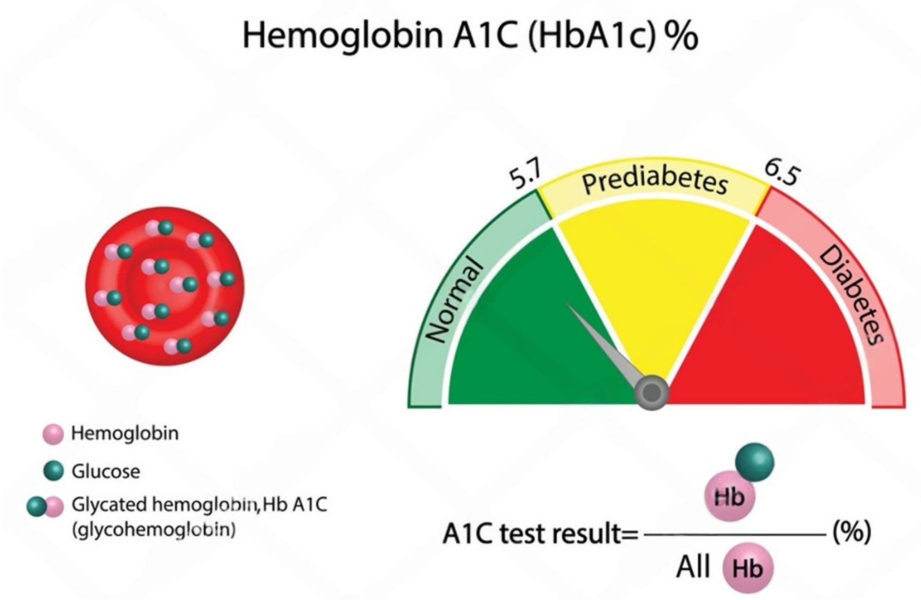

Figure 1.1: Image by Shutterstock. Source: `https://tinyurl.com/2nvxs5nz`

2 Data Description and Analysis Problem

2.1 Data Source

The data for this project is the National Health and Nutrition Examination Survey (NHANES) GH data, which was collected by the U.S Centers for Disease Control and Prevention. The NHANES examines participants' health via interviews, evaluations, and laboratory results. Participants for the survey are of many different age groups, and was randomly selected to represents the diverse communities of the United States; thus, providing a good representation of the public health status [1].

2.2 Previous Uses of the Data and This Project's Analysis

The NHANES GH data been used in various settings, but usually in tandem with other datasets. An example is a study investigating the relationship between sleep symptoms and GH [2], or a study examining the relationship between hemoglobin glycation index and mortality [3]. We have noticed a study on Kaggle also utilising the same dataset; however, this study aims to predict types of diabetes using various machine learning and deep learning method, which differs greatly from our project [4]. The study with the closest use of data to our project is a research on the link between waist circumference and type 2 diabetes; however, in terms of methodology, the machine learning models logistic regression and linear regression were used, and not Bayesian models like in this project [5].

Therefore, the use of Bayesian statistics and modeling in this particular topic has made sure that this project is not an iteration of other analyses, but a meaningful exploration on its own.

2.3 Variables and Structure of the Data

The following variables are included in the dataset:

- **seqn**: respondent sequence number
- **sex**: recorded as male and female
- **age**: recorded in years
- **re**: race/ ethnicity
- **income**: grouped into income groups
- **tx**: if person is on insulin/ diabetes medicines
- **dx**: if person is diagnosed with diabetes mellitus or pre-diabetic
- **wt**: weight in kilograms (kg)
- **ht**: standing height in centimeters (cm)
- **bmi**: Body Mass Index (BMI)
- **leg**: upper leg length in cm
- **arml**: upper arm length in cm
- **armc**: arm circumference in cm
- **waist**: waist circumference in cm
- **tri**: triceps skinfold in millimeters (mm)
- **sub**: subscapular skinfold in mm
- **gh**: glycohemoglobin as a percentage
- **albumin**: albumin in g/dL
- **bun**: blood urea nitrogen in mg/dL
- **SCr**: creatinine in mg/dL

In this project, we want to utilise measurements that would be easy for the general public to record on their own, while also ensuring that these measures have strong scientific evidence of being predictors or risk factors of diabetes. The motivation behind keeping self-measurements simple is to promote more frequent and proactive monitoring for early detection of health decline. BMI and waist circumference are well-proven predictors of diabetes, with waist circumference having a larger positive association with diabetic risks [6]. This is particularly stronger for individuals whose BMI are at a low or normal level [7]. Because of this, waist circumference is used as a parameter for this project's Bayesian models. Age is also a key predictor that is connected to various diabetic risk factors, and thus, was also included in our models [8].

3 Models

3.1 Intercept Baseline Model

This model serves as a simple baseline against which the more complex non-hierarchical and hierarchical models can be compared. By predicting GH using only an intercept, the model estimates the overall population mean while ignoring individual-level predictors. This provides a reference point for evaluating whether later models, which incorporate covariates or group-level variation, offer meaningful improvements in fit or predictive performance. We use a Student-t likelihood to allow for heavier tails, specify weakly informative priors on both the intercept and residual scale, and run standard MCMC settings to ensure stable posterior sampling.

Model Specification

```
intercept_formula <- bf(
  gh ~ 1,
  family = student(),
  center = FALSE
)
```

Priors

```
intercept_priors <- c(
  prior(normal(5, 2), class = "b", coef = "Intercept"),
  prior(exponential(1), class = "sigma")
)
```

brms Code

```
fit_intercept <- brm(
  formula = intercept_formula,
  prior = intercept_priors,
  data = nhgh_model,
  chains = 4,
  iter = 2000,
  warmup = 1000
)
```

MCMC Settings The model is fit using four chains, each with 2,000 iterations and a 1,000-iteration warmup phase, providing a total of 4,000 post-warmup samples. This configuration ensures sufficient mixing and effective sample size for a stable baseline estimate.

Convergence Diagnostics

```
Family: student
  Links: mu = identity
Formula: gh ~ 1
   Data: nhgh_model (Number of observations: 6264)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept     5.45      0.01     5.44     5.46 1.00     2940     2816
```

```
Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.34      0.01     0.33     0.35 1.00     2218     2273
nu        1.85      0.05     1.74     1.95 1.00     2178     2522

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

[1] "Divergence of the model: 0"
```

The baseline model has $\hat{R} = 1.00$, meaning that the model has converged. This indicates that the model and the estimates for the parameters are reliable. Both of the bulk and tail Effective Sample Size (ESS) are all larger than 2000, meaning that the posterior distribution has been well explored, and thus, the inference is reliable. As there is no divergence, we can conclude that the HMC algorithm has explored the posterior distribution without any issues. The ν value is 1.85, signifying the heavy-tail mentioned above. As normal glycohemoglobin levels is below 5.7% [9], from the estimate of the intercept being 5.45% and 95%CI = [5.44, 5.46], it seems most individuals in this dataset has a healthy level of glycohemoglobin.

3.2 Non-Hierarchical Model

This non-hierarchical model extends the baseline by introducing individual-level predictors: standardized age, standardized waist-to-height ratio, and race indicators. The motivation for this specification is to evaluate whether these covariates meaningfully improve the prediction of glycohemoglobin (GH) beyond the population-level mean. Including these predictors allows us to quantify their direct effects while still assuming that all individuals share a single underlying regression structure. As before, we use a Student-t likelihood to account for potential outliers and specify weakly informative priors that regularize the coefficients without imposing strong assumptions. This model therefore serves as a critical comparison point for the later hierarchical models, which will allow these relationships to vary across income and racial groups.

Model Specification

```
nonhier_formula <- bf(
gh ~ 1 + age_z + waist_z + race,
family = student()
)
```

Priors

```
nonhier_priors <- c(
prior(normal(5, 2), class = "Intercept"),
prior(normal(0, 1), class = "b", coef = "age_z"),
prior(normal(0, 1), class = "b", coef = "waist_z"),
prior(normal(0, 1), class = "b"), # covers race dummy coefficients
prior(exponential(1), class = "sigma")
)
```

brms Code

```
fit_nonhier <- brm(
formula = nonhier_formula,
prior = nonhier_priors,
data = nhgh_model,
chains = 4,
iter = 2000,
warmup = 1000
)
```

MCMC Settings The model is run using four chains with 2,000 iterations each and a 1,000-iteration warmup. This setup yields 4,000 post-warmup samples, providing sufficient effective sample size to assess the contribution of individual-level predictors before progressing to hierarchical model structures.

Convergence Diagnostics

```
    Family: student
  Links: mu = identity
Formula: gh ~ 1 + age_z + waist_z + race
   Data: nhgh_model (Number of observations: 6264)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
                                   Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept                              5.53      0.01     5.51     5.55 1.00     2729     2886
age_z                                  0.19      0.01     0.18     0.20 1.00     5002     3209
waist_z                                0.08      0.01     0.07     0.09 1.00     6058     3142
raceOtherHispanic                     -0.02      0.02    -0.05     0.02 1.00     3605     2882
raceNonMHispanicWhite                 -0.11      0.01    -0.14    -0.09 1.00     2671     3152
raceNonMHispanicBlack                  0.11      0.02     0.08     0.14 1.00     3706     3088
raceOtherRaceIncludingMultiMRacial     0.04      0.02    -0.00     0.08 1.00     4201     3428

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.28      0.00     0.27     0.29 1.00     4256     3362
nu        1.77      0.05     1.68     1.87 1.00     4051     3354

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

[1] "Divergence of the model: 0"
```

We can see that in our model, the $\hat{R}$ values are all equal to 1.00, meaning that across all parameters, the model has properly converged. This indicates that the model and the estimates for the parameters are reliable. Both of the bulk and tail Effective Sample Size (ESS) are all larger than 2000, meaning that the posterior distribution has been well explored, and thus, the inference is reliable. As there is no divergence, we can conclude that the HMC algorithm has explored the posterior distribution without any issues.

The estimate for the standardised age parameter is noticeably larger than that of the standardised waist circumference parameter, suggesting that age is a stronger predictor for glycohemoglobin. The reference race group in this case is "Mexican American". The racial group "Non-Hispanic White" has a notably lower predicted level of glycohemoglobin compared to the reference group, while that of the group "Non-Hispanic Black" is notably higher. For the other groups, the estimated levels are similar to the reference group with no clear evidence of difference, as their 95% Credible Interval (CI) includes 0.

3.3 Hierarchical Model: Income Variation

This model introduces a varying intercept structure for income groups, allowing the baseline level of glycohemoglobin (GH) to differ across socioeconomic categories. The motivation for this model is to capture potential disparities linked to healthcare access, lifestyle, and structural factors correlated with income. By keeping age and waist-to-height ratio as fixed effects while allowing income-level intercepts to vary, the model examines whether socioeconomic status meaningfully moderates overall GH levels. As before, a Student- t t likelihood is used for robustness, and weakly informative priors constrain the group-level variation to plausible ranges.

Model Specification

```
hier_formula_inc <- bf(
gh ~ 1 + age_z + waist_z + (1 | income),
family = student(),
center = FALSE
)
```

Priors

```
hier_priors_inc <- c(
prior(normal(5, 2), class = "b", coef = "Intercept"),
prior(normal(0, 1), class = "b", coef = "age_z"),
prior(normal(0, 1), class = "b", coef = "waist_z"),

prior(exponential(1), class = "sd", group = "income", coef = "Intercept"),
prior(exponential(1), class = "sigma")
)
```

brms Code

```
fit_hier_inc <- brm(
formula = hier_formula_inc,
prior = hier_priors_inc,
data = nhgh_model,
chains = 4,
iter = 2000,
warmup = 1000,
control = list(adapt_delta = 0.99)
)
```

MCMC Settings This model is run with four chains, 2,000 iterations per chain, and a 1,000-iteration warmup. A higher *adapt_delta* is used to ensure stable sampling for hierarchical parameters.

Convergence Diagnostics

```
    Family: student
  Links: mu = identity
Formula: gh ~ 1 + age_z + waist_z + (1 | income)
   Data: nhgh_model (Number of observations: 6264)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~income (Number of levels: 14)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.03      0.01     0.01     0.05 1.00     1474     1716

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept     5.49      0.01     5.48     5.51 1.00     1540     2104
age_z         0.18      0.01     0.17     0.19 1.00     5408     2741
waist_z       0.08      0.01     0.06     0.09 1.00     5574     3378

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.29      0.00     0.28     0.30 1.00     3611     3061
nu        1.82      0.05     1.72     1.93 1.00     3767     2835

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

[1] "Divergence of the model: 0"
```

We can see that in our model, the $\hat{R}$ values are all equal to 1.00, meaning that across all parameters, the model has properly converged. This indicates that the model and the estimates for the parameters are reliable. Both of the bulk and tail ESS are all larger than 1000, meaning that the posterior distribution has been well explored, and thus, the inference is reliable. As there is no divergence, we can conclude that the HMC algorithm has explored the posterior distribution without any issues.

The estimate for the standard deviation of the income intercept is very small at 0.03, with 95% CI at [0.01, 0.05], suggesting that different income levels only account for very limited variability for glycohemoglobin prediction.

3.4 Hierarchical Model: Race Variation

This model allows both intercepts and slopes for age and waist-to-height ratio to vary across racial groups. The motivation is to test whether physiological predictors differ in strength or direction across populations that experience distinct structural, genetic, or environmental influences. Allowing varying slopes enables the model to capture heterogeneity in how GH relates to age or central adiposity across race categories. Weakly informative exponential priors on the group-level standard deviations and an LKJ prior on the correlation matrix help regularize the hierarchical structure while allowing meaningful variation.

Model Specification

```
hier_formula_race <- bf(
gh ~ 1 + age_z + waist_z + (1 + age_z + waist_z | race),
family = student(),
center = FALSE
)
```

Priors

```
hier_priors_race <- c(
prior(normal(5, 2), class = "b", coef = "Intercept"),
prior(normal(0, 1), class = "b", coef = "age_z"),
prior(normal(0, 1), class = "b", coef = "waist_z"),

prior(exponential(1), class = "sd", group = "race", coef = "Intercept"),
prior(exponential(1), class = "sd", group = "race", coef = "age_z"),
prior(exponential(1), class = "sd", group = "race", coef = "waist_z"),
prior(lkj(2), class = "cor", group = "race"),

prior(exponential(1), class = "sigma")
)
```

brms Code

```
fit_hier_race <- brm(
formula = hier_formula_race,
prior = hier_priors_race,
data = nhgh_model,
chains = 4,
iter = 2000,
warmup = 1000,
control = list(adapt_delta = 0.99)
)
```

MCMC Settings Four chains, 2,000 iterations, and a 1,000-iteration warmup are used, with an increased *adapt_delta* to accommodate the more complex correlation structure among varying slopes.

Convergence Diagnostics

```
Family: student
  Links: mu = identity
Formula: gh ~ 1 + age_z + waist_z + (1 + age_z + waist_z | race)
   Data: nhgh_model (Number of observations: 6264)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~race (Number of levels: 5)
                       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)              0.13      0.08     0.06     0.34 1.01     1319     1387
sd(age_z)                  0.04      0.03     0.01     0.12 1.00     1457     2192
sd(waist_z)                0.03      0.03     0.00     0.10 1.00      896      949
cor(Intercept,age_z)       0.05      0.36    -0.64     0.69 1.00     3439     2775
cor(Intercept,waist_z)     0.16      0.38    -0.60     0.82 1.00     3657     2964
cor(age_z,waist_z)        -0.13      0.39    -0.80     0.64 1.00     3258     2971

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept     5.54      0.07     5.40     5.68 1.01     1143     1305
age_z         0.20      0.02     0.15     0.25 1.00     1677     1769
waist_z       0.08      0.02     0.03     0.11 1.00     1372      869

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.28      0.00     0.27     0.29 1.00     4286     3387
nu        1.77      0.05     1.67     1.87 1.00     4321     3185

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

[1] "Divergence of the model: 1"
```

We can see that in our model, the $\hat{R}$ values are all ~ 1.0, meaning that across all parameters, the model has properly converged. This indicates that the model and the estimates for the parameters are reliable. Both of the bulk and tail ESS are all larger than 800, meaning that the posterior distribution has been well explored, and thus, the inference is reliable. As there is 1 divergence, the HMC algorithm has encountered a problematic area when exploring the posterior distribution. However, this does not seem to be a big concern.

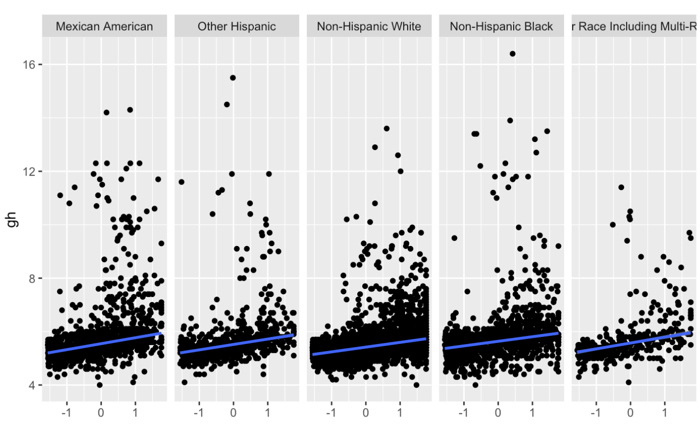

Figure 3.1: Conditional effect plot for age

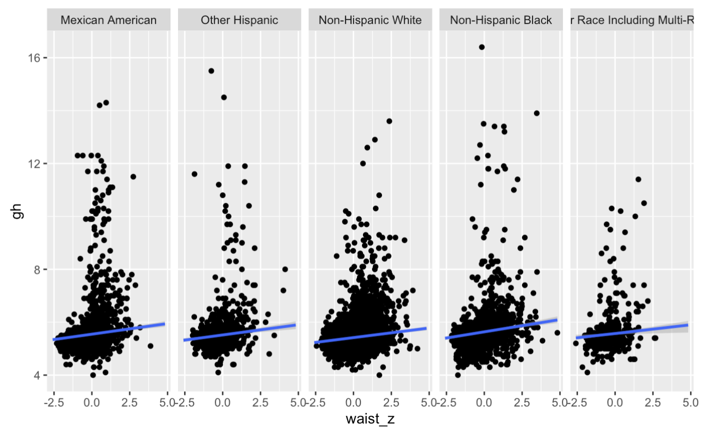

Figure 3.2: Conditional effect plot for waist circumference

The model includes varying intercepts by race. From the diagnostics, we have the standard deviation between the races as 0.13, with the 95% CI as [0.06, 0.34]. This means there there is a difference between how the model predict glycohemoglobin for each race; however, the difference is not large. From the diagnostics values and the plots 3.1 and 3.2, there is substantial overlapping of the posterior distributions for the varying slopes of age and waist circumference, thus there is no clear evidence that one predictor varies more by race than the other. Across the racial groups, both predictors exhibit rather small slope variability.

3.5 Hierarchical Model: Income and Race Variation

This combined hierarchical model includes varying intercepts for income groups and varying intercepts and slopes for racial groups. Its purpose is to jointly examine socioeconomic and racial disparities in glycohemoglobin, capturing both baseline differences and differential effects of age and waist-to-height ratio. By allowing multiple layers of population structure, this model tests whether predictive patterns vary across intersecting social categories—an important step in identifying how early markers of dysregulated glucose may differ across the population.

Model Specification

```
hier_formula_inc_race <- bf(
gh ~ 1 + age_z + waist_z +
(1 + age_z + waist_z | race) +
(1 | income),
family = student(),
center = FALSE
)
```

Priors

```
hier_priors_inc_race <- c(
prior(normal(5, 2), class = "b", coef = "Intercept"),
prior(normal(0, 1), class = "b", coef = "age_z"),
prior(normal(0, 1), class = "b", coef = "waist_z"),

prior(exponential(1), class = "sd", group = "race", coef = "Intercept"),
prior(exponential(1), class = "sd", group = "race", coef = "age_z"),
prior(exponential(1), class = "sd", group = "race", coef = "waist_z"),
prior(lkj(2), class = "cor", group = "race"),
```

```
prior(exponential(1), class = "sd", group = "income", coef = "Intercept"),
prior(exponential(1), class = "sigma")
)
```

brms Code

```
fit_hier_inc_race <- brm(
formula = hier_formula_inc_race,
prior = hier_priors_inc_race,
data = nhgh_model,
chains = 4,
iter = 2000,
warmup = 1000,
control = list(adapt_delta = 0.99)
)
```

MCMC Settings As with the other hierarchical models, four chains with 2,000 iterations and a 1,000-iteration warmup are used. The higher *adapt_delta* again helps ensure reliable sampling in the presence of correlated group-level effects.

Convergence Diagnostics

```
 Family: student
  Links: mu = identity
Formula: gh ~ 1 + age_z + waist_z + (1 + age_z + waist_z | race) + (1 | income)
   Data: nhgh_model (Number of observations: 6264)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~income (Number of levels: 14)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.01      0.01     0.00     0.03 1.00     1237     1865

~race (Number of levels: 5)
                       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)              0.13      0.08     0.05     0.34 1.00     1585     1810
sd(age_z)                  0.05      0.03     0.01     0.13 1.00     1044     1480
sd(waist_z)                0.03      0.03     0.00     0.11 1.00      670      601
cor(Intercept,age_z)       0.05      0.36    -0.64     0.72 1.00     3478     2890
cor(Intercept,waist_z)     0.17      0.38    -0.60     0.81 1.00     4124     3190
cor(age_z,waist_z)        -0.14      0.40    -0.82     0.68 1.00     3604     3105

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept     5.54      0.07     5.40     5.68 1.00     1262     1728
age_z         0.20      0.03     0.15     0.25 1.00     1355     1509
waist_z       0.08      0.02     0.04     0.11 1.00     1096      524

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.28      0.00     0.27     0.29 1.00     5318     3055
nu        1.77      0.05     1.67     1.87 1.00     5512     3020

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

[1] "Divergence of the model: 0"
```

We can see that in our model, the $\hat{R}$ values are all equal to 1.00, meaning that across all parameters, the model has properly converged. This indicates that the model and the estimates for the parameters are reliable. Both of the bulk and tail ESS are all larger than 500, meaning that the posterior distribution has been well explored, and thus, the inference is reliable. As there is no divergence, we can conclude that the HMC algorithm has explored the posterior distribution without any issues.

From the convergence diagnostics, it is clear that there is small intercept variability by race (estimate of the standard deviation = 0.13, 95% CI = [0.05, 0.34]) and very little to no intercept variability by income (estimate of the standard deviation = 0.01, 95% CI = [0.00, 0.03]). This suggests that income affects glycohemoglobin level to a very limited degree, while race is a more notable factor. Across the racial groups, there is substantial overlapping of the posterior distributions for the varying slopes of age and waist circumference, thus there is no clear evidence that one predictor varies more by race than the other: both predictors exhibit rather small slope variability.

3.6 Posterior Predictive Checks

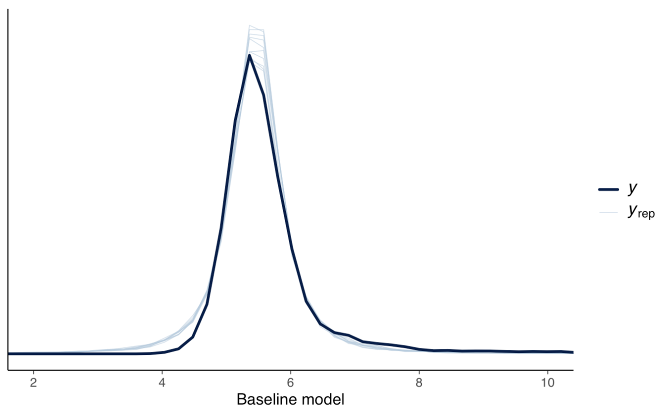

Figure 3.3: Posterior predictive check for the baseline model

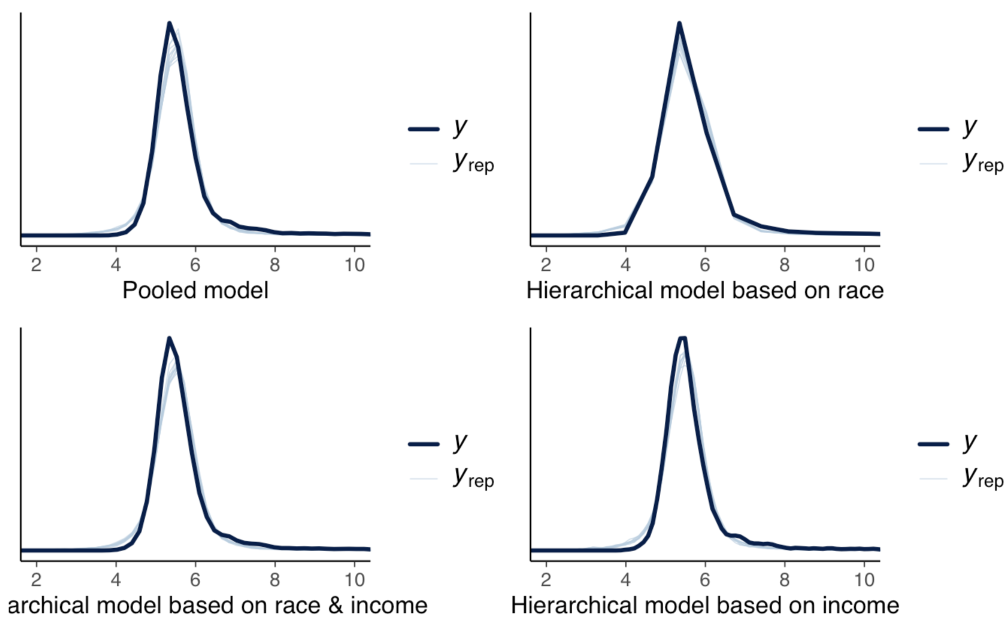

Figure 3.4: Posterior predictive check for each predictive model

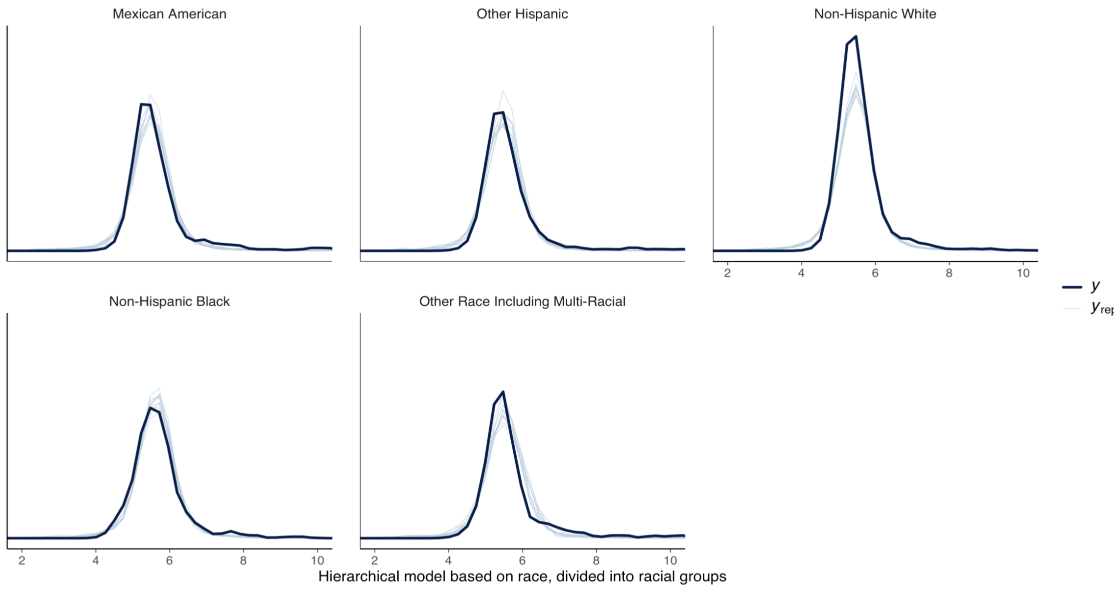

Figure 3.5: Posterior predictive check for the hierarchical model based on race

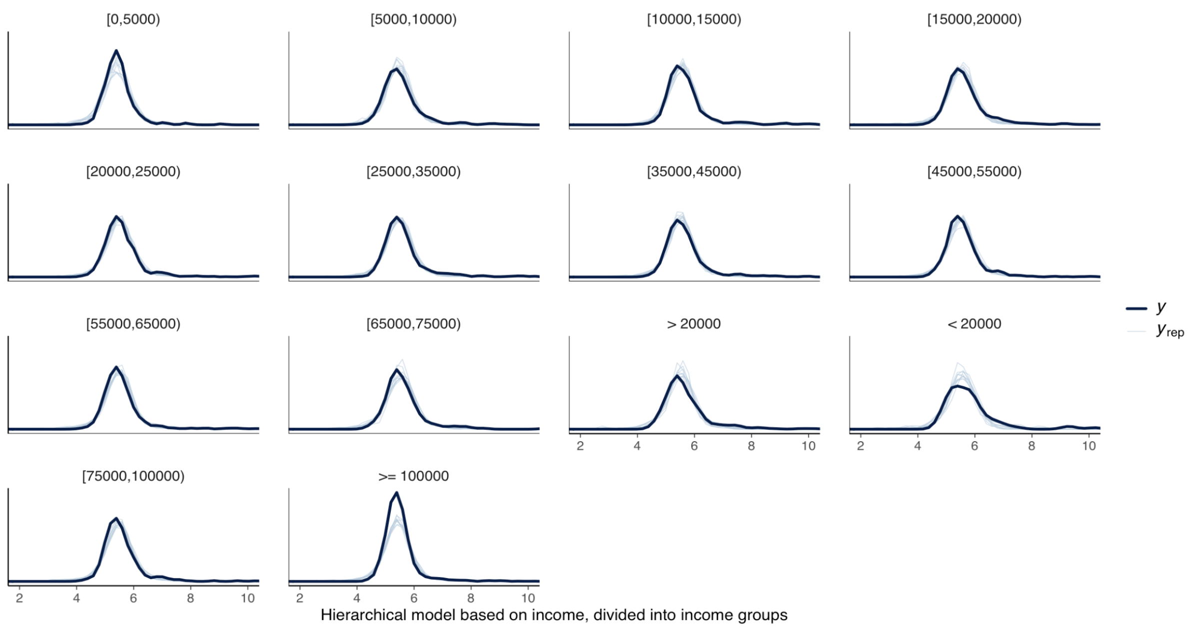

Figure 3.6: Posterior predictive check for the hierarchical model based on income

We can see that for all predictive 4 models, the predicted data graphs seem to be having a highly similar distribution to the observed data, and performs better than the intercept-only baseline model. This high degree of similarity is preserved when examining the distributions for each racial group. When distribution is examined based on racial groups, the "Non-Hispanic White" group has notably less accurate prediction. When the distribution is examined based on income, there are some income groups whose predictions are less accurate, such as the groups "$<$ 20000" and "$>=$ 100000". However, for the group "$<$ 20000" in particular, this is likely due to its very small sample size of 75 people. This small sample size can lead to more noise in the observed distributions. Overall, we see that the posterior predictive check shows that the models performs well, with some subgroups having limited accuracy in terms of predictive performance.

3.7 Sensitivity Analysis

We evaluated the sensitivity of all five models to prior and likelihood perturbations using power-scaling methods, implemented via *priorsense* R package. This approach quantifies how much key posterior quantities change when the effective strength of the priors or likelihoods is artificially increased or decreased. This method allows us to detect prior-data conflict, weak identifiability, or parameters whose esimtates depend strongly on the chosen model components.

Given that priorsense provides two complementary forms of sensitivity assessment, we used both:

- Local sensitivity (used for numerical diagnostics): We applied local power-scaling using $\alpha \in [0.99, 1.01]$ in order to compute CJS distances for each parameter. These distances quantify how rapidly posterior expectations change under infinitesimal preturbations of the prior or likelihood. This is the basis for the sensitivity tables reported below.

- Global robustness checks (used for plots): For visual diagnostics, we used *priorsense's* global power-scaling grid: $\alpha \in \{0.8, 1.0, 1.25\}$. These larger perturbations produce posterior density ovelays and posterior mean/standard eviation trajectories across α, which more clearly reveal whether posterior inferences remain stable when the model is sustantially perturbed.

3.7.1 Baseline Intercept Model

The intercept model exhibits negligible prior sensitivity for all parameters, with CJS distance ≤ 0.003. Likelihood sensitivity was modest $\approx 0.09 - 0.11$, consistent with the behavior of Student-t distributional parameters. No prior-data conflicts were detected.

Global plots confirm that the posterior for the intercept barely moves across $\alpha = 0.8 - 1.25$. This high stability arises from the large sample size $N = 6264$, which dominates both prior and likelihood perturbations. Despite its stability, this model dramatically underfits the data (see Section 4).

Parameter	**CJS (prior)**	**CJS (likelihood)**	**Diagnosis**
b_Intercept	0.000	0.098	–
σ	0.001	0.091	–
ν	0.003	0.109	–

Table 3.1: Local power-scaling sensitivity diagnostics for the intercept-only model. Values are cumulative Jensen–Shannon distances (CJS) under small perturbations $\alpha \in [0.99, 1.01]$.

3.7.2 Non-Hierarchical Model

The non-hierarchical model shows very low sensitivity for all regression coefficients, including age, waist, and the race variables. Prior sensitivity was essentially zero, and likelihood sensitivity remained very small $(0.08 - 0.11)$.

Global density overlays show nearly perfect overlap for $\alpha = 0.8 - 1.25$, indicating that fixed effects are highly stable. Furthermore, no prior-data conflicts arise. These results indicate that the fixed-effect regression parameters are well-identified by the data, and the model's conclusions are robust to moderate misspecifications of either prior or likelihood strength.

Parameter	CJS (prior)	CJS (likelihood)	Diagnosis
b_Intercept	0.000	0.086	–
b_age_z	0.000	0.082	–
b_waist_z	0.000	0.080	–
b_raceOtherHispanic	0.000	0.084	–
b_raceNonMHispanicWhite	0.000	0.093	–
b_raceNonMHispanicBlack	0.000	0.097	–
b_raceOtherRaceIncludingMultiMRacial	0.000	0.092	–
σ	0.001	0.108	–
ν	0.003	0.095	–

Table 3.2: Local power-scaling sensitivity diagnostics for the non-hierarchical regression model. Values are CJS distances for $\alpha \in [0.99, 1.01]$.

3.7.3 Hierarchical Model: Income Variation

For the income-only hierarchical model, fixed effects remain very stable, with minimal local sensitivity and overlapping density curves across global α. The income-level intercept standard deviation shows modest likelihood sensitivity, reflecting the fact that hierarchical variance parameters are typically less precisely identified, especially with small group-level variability (as observed in this model).

However, both local CJS distances and global plots show that these deviations are small in absolute magnitude and do not meaningfully affect inference. No prior–data conflict is detected.

Parameter	CJS (prior)	CJS (likelihood)	Diagnosis
b_Intercept	0.001	0.028	–
b_age_z	0.000	0.080	–
b_waist_z	0.000	0.098	–
sd_income__Intercept	0.002	0.093	–
σ	0.001	0.095	–
ν	0.003	0.075	–

Table 3.3: Local power-scaling sensitivity diagnostics for the hierarchical model with varying income intercepts. Values are CJS distances for $\alpha \in [0.99, 1.01]$.

3.7.4 Hierarchical model: Race Variation

This model exhibits marginally higher sensitivity compared to the other models. Fixed effects (b_age_z, b_waist_z) again show very low sensitivity. However, several group-level variance components, particularly sd_race_Intercept, exhibit moderate sensitivity under likelihood perturbations, refelcted in slightly elavated CJS distances and the "potential prior-data conflict" flag in local diagnostics.

This behavior is expected because the model estimates multiple random slopes with only five race groups, which inherently limits identifiability. Importantly, absolute changes remain small, and both density overlays and posterior mean/standard deviation trajectories indicate overall stability. Thus, while the hierarchical variance components are somewhat sensitive, the model remains well-behaved.

Parameter	**CJS (prior)**	**CJS (likelihood)**	**Diagnosis**
b_Intercept	0.010	0.034	–
b_age_z	0.007	0.023	–
b_waist_z	0.006	0.036	–
sd_race__Intercept	0.057	0.052	potential prior–data conflict
sd_race__age_z	0.029	0.076	–
sd_race__waist_z	0.023	0.189	–
σ	0.004	0.103	–
ν	0.004	0.107	–

Table 3.4: Local power-scaling sensitivity diagnostics for the hierarchical model with varying intercepts and slopes by race. Values are CJS distances for $\alpha \in [0.99, 1.01]$.

3.7.5 Hierarchical Model: Race and Income Variation

This model displays a similar sensitivity pattern to the hierarchical model with varying race. Fixed effects (b_age_z, b_waist_z, and the intercept) show very low sesitivity under both prior and likelihood perturbations. Several race-level variance components, particularly sd_race_Intercept and sd_race__age_z, exhibit moderate sensitivity, reflected in elevated CJS distances and flags indicating potential prior–data conflict. This is expected given the small number of race groups and the estimation of multiple varying effects.

The income-level intercept variance shows no conflict, despite a higher likelihood sensitivity value, because the underlying variance is extremely small and thus only weakly informed by the data. Global density overlays and posterior mean/standard deviation trajectories across α confirm that all parameters remain stable and well-behaved. Overall, this model demonstrates robust posterior behavior, with sensitivity only in the hierarchical variance parameters where limited identifiability is expected.

4 Model Comparison

4.1 PSIS-LOO Cross-Validation

We compared predictive accuracy using Pareto-smoothed importance sampling Leave-One-Out cross-validation (PSIS-LOO). All models had good Pareto k values $k < 0.7$, indicating reliable LOO estimates. The elpd_loo, elpd difference relative to the best model values, and the standard error of the difference are reported in Table 4.1.

4.2 Interpretation of Comparison Results

We draw the following conclusions based on the model comparison results:

Parameter	**CJS (prior)**	**CJS (likelihood)**	**Diagnosis**
b_Intercept	0.012	0.021	–
b_age_z	0.013	0.032	–
b_waist_z	0.008	0.046	–
sd_race__Intercept	0.062	0.038	potential strong prior / weak likelihood
sd_race__age_z	0.058	0.090	potential prior–data conflict
sd_race__waist_z	0.035	0.140	–
sd_income__Intercept	0.004	0.234	–
σ	0.004	0.084	–
ν	0.007	0.086	–

Table 3.5: Local power-scaling sensitivity diagnostics for the hierarchical model with varying race and income effects. Values are CJS distances for $\alpha \in [0.99, 1.01]$.

Model	**elpd_loo**	**Δelpd**	**SE(Δelpd)**
Hierarchical: Race (best)	−4766.6	0.0	0.0
Hierarchical: Race + Income	−4767.3	−0.6	1.3
Non-hierarchical	−4772.3	−5.7	4.6
Hierarchical: Income only	−4900.6	−134.0	18.9
Intercept-only	−5782.9	−1016.2	42.0

Table 4.1: Summary of PSIS-LOO predictive performance for all fitted models. Higher elpd_loo indicates better predictive accuracy. The hierarchical race model is used as the reference model.

- **Hierarchical structure improves predictive accuracy**: The models with highest predictive accuracy were the hierarchical (race variation) and the hierarchical (race and income variation) models. These two models are statistically indistinguishable ($\Delta\text{elpd} = -0.6 \pm 1.3$), meaning that adding income-level variation does not improve predictive performance beyond race grouping.

- **Non-hierarchical model performs moderately worse**: The non-hierarchical model performs worse than either hierarchical race model ($\Delta\text{elpd} \approx -6$), but this difference is small relative to standard error, implying that hierarchical partial pooling over race provides a modest but consistent improvement.

- **Income-only hierarchical structure is not helpful**: The hierarchical income model performs significantly worse ($\Delta\text{elpd} \approx -134$). This reflects that income-level intercept variability is extremely small and does not contribute meaningful structure for prediction.

- **The baseline intercept model performs significantly poorly**: The baseline model underfits the data by a massive margin ($\Delta\text{elpd} \approx -1016$). This confirms that age, waist circumference, and race are essential predictors of glycohemoglobin.

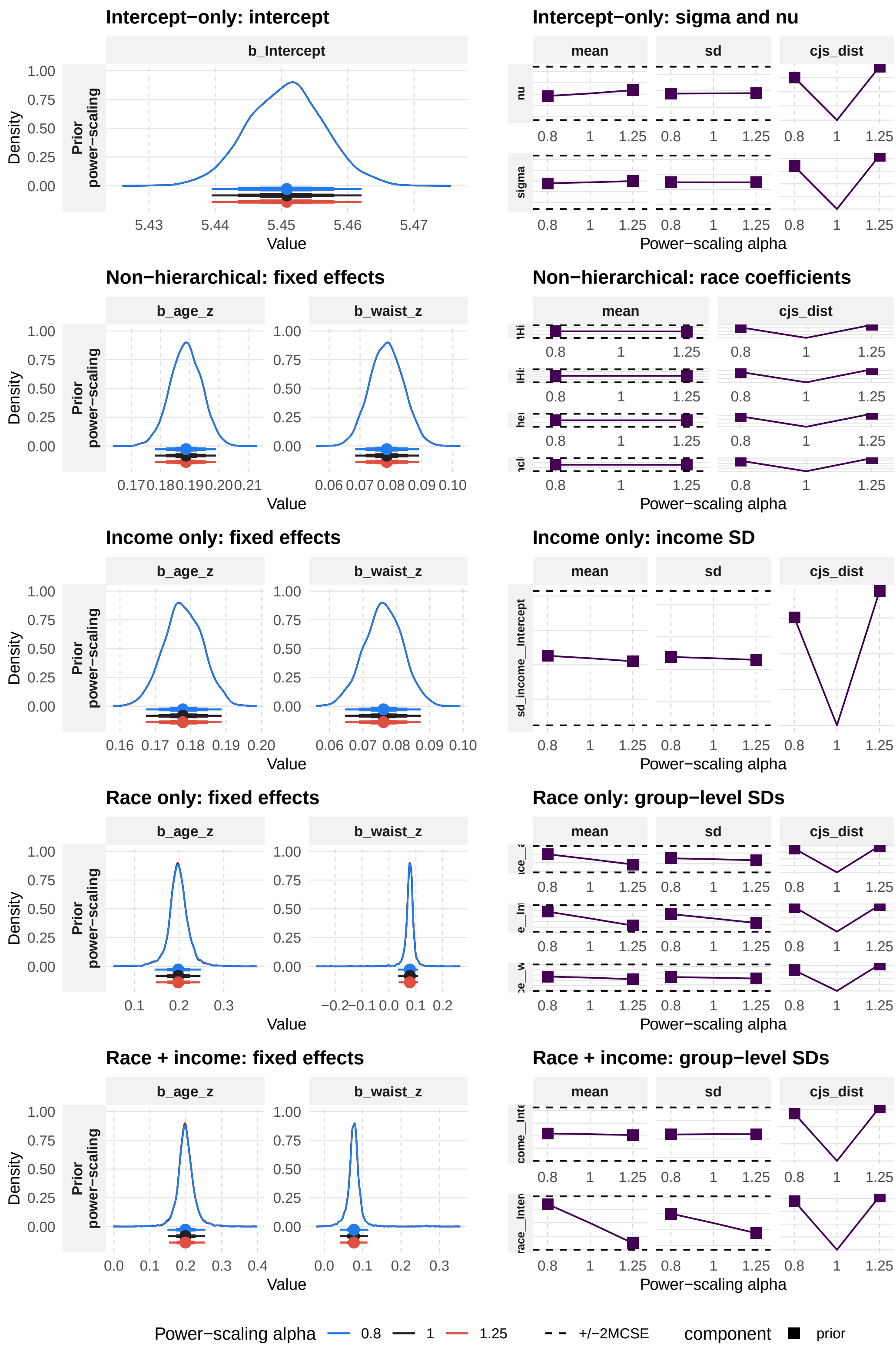

Figure 3.7: Power-scaling sensitivity analysis for all five models. Left column: posterior densities of selected fixed effects under prior power-scaling. Right column: posterior mean and standard deviation of selected variance and distributional parameters as a function of α.

5 Issues and Potential Improvements

For the hierarchical model with race variation, there is 1 divergence, even with adapt_delta fixed to 0.99. This does not meaningfully compromise the reliability of the inference, as the model's $\hat{R}$ values are all all ~ 1.0. However, to improve this, we can fix the adapt_delta to a value closer to 1, such as 0.999, and fit the model again.

From the posterior predictive checks, for the hierarchical model with race variation, the racial group "Non-Hispanic White" shows worse predictive performance compared to other groups, while for the hierarchical model with income variation, this is the case with the groups "< 20000" and "$>= 100000$". Since the group "< 20000" has a very small sample size of 75 people, the less accurate prediction is not likely due to the model and can be improved by binning it together with other income groups that is in the range [0, 20000). A possible cause that leads to less accurate predictive performance is a high variability within the group. To fix this, we could inspect the groups for possible outliers, and also adjust the model to allow the value of σ to vary across the groups.

6 Conclusion

6.1 What Was Learned from the Data Analysis

Across all predictive models presented in this project, it is clear that level of glycohemoglobin can be well predicted using age and waist circumference. The hierarchical model with race variation has converged well, producing reliable inference, with small variability across the various racial groups, suggesting that the groups have different ranges of glycohemoglobin level. However, as the slopes of age and waist circumference have substantial overlapping across the different groups, there is no clear evidence that the predictor strengths vary significantly. For the hierarchical model with income variation, in this dataset, income does not contribute meaningfully to the model. In the hierarchical model with both race and income variation, there is small intercept variability by race, but very limited intercept variability by income.

After model comparison, we see that the baseline model performs most poorly, signifying that age and waist circumference are essential in the prediction of glycohemoglobin. The hierarchical model with race variation, together with the hierarchical model with race and income variation performs the best with statistically indistinguishable performance from the race-only model, indicating that combining income level variation does not improve prediction. The non hierarchical model does not perform as good as the 2 race-based models; however, its predictive performance follows closely, suggesting that there is small advantage to introduce hierarchical structure using racial groups. The hierarchical model with only income variation performs notably worse than the other hierarchical models, suggesting that intercept variability at income level does not meaningfully contribute to predictive performance.

We can conclude that while introducing a hierarchical structure with race variation provide small but meaningful improve in predictive performance, the main predictors of waist circumference and age remain the same across the groups. This means that early detection effort should focus on these strong predictors, and incorporate racial group variations if the information is readily available.

6.2 Self-Reflection on Project Process

This project has allowed our group to better understand and familiarise ourselves with how to approach an analysis question utilising the principles of Bayesian Data Analysis. A key learning point that has set working on the project aside from simply doing the assignments on MyCourses is that the project urges us to interpret our analysis results more critically, as well as consider what this would imply for the research question. Thus, this bridges the gap between knowing the knowledge and using the knowledge in a meaningful way. Thanks to this project, we have improved our technical skills in conducting Bayesian Data Analysis, and greatly refined our ability to interpret and draw conclusions from the results.

7 References

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A Appendix

The code that was written for this project is included here.

Loading library and data:

```
if (!require(tidybayes)) {
    install.packages("tidybayes")
    library(tidybayes)
}
library(dplyr)
if (!require(brms)) {
    install.packages("brms")
```

```
    library(brms)
}
if(!require(ggplot2)) {
  install.packages("ggplot2")
  library(ggplot2)
}
if (!require(metadat)) {
  install.packages("metadat")
  library(metadat)
}
if(!require(cmdstanr)){
    install.packages("cmdstanr", repos = c("https://mc-stan.org/r-packages/", getOption("repos")))
    library(cmdstanr)
}
cmdstan_installed <- function(){
  res <- try(out <- cmdstanr::cmdstan_path(), silent = TRUE)
  !inherits(res, "try-error")
}
if(!cmdstan_installed()){
    install_cmdstan()
}
library(loo)

if (!require(gridExtra)) {
  install.packages("gridExtra")
  library(gridExtra)
}

if (!require(priorsense)) {
  install.packages("priorsense")
  library(priorsense)
}


load("nhgh.rda")


nhgh_model <- nhgh |>
  dplyr::select(gh, age, race = re, waist, income) |>
  tidyr::drop_na()

nhgh_model <- nhgh_model |>
  mutate(
    age_z   = as.numeric(scale(age)),
    waist_z = as.numeric(scale(waist))
  )

summary(nhgh_model$gh)
sd(nhgh_model$gh)
# use a N(5, 2) prior.
#weakly informative centered around the true data mean, slightly higher SD to account for outliers.
hist(nhgh_model$gh, breaks = 40, col = "gray")
```

Fitting and looking at convergence diagnostics for the baseline model:

```
intercept_formula <- bf(
  gh ~ 1,
  family = "student",
  center = FALSE
)

intercept_priors <- c(
  prior(normal(5, 2), class = "b", coef = "Intercept"),
  prior(exponential(1), class = "sigma")
)

fit_intercept <- brm(
  formula = intercept_formula,
  prior = intercept_priors,
  data = nhgh_model,
  chains = 4,
```

```
  iter = 2000,
  warmup = 1000,
)

summary(fit_intercept)

nutsparams_fit_intercept <- nuts_params(fit_intercept)

sprintf("Divergence of the model: %i", sum(subset(nutsparams_fit_intercept, Parameter == "divergent__")$Value))
```

Fitting and looking at convergence diagnostics for the non-hierarchical model:

```
nonhier_formula <- bf(
  gh ~ 1 + age_z + waist_z + race,
  family = student()
)

nonhier_priors <- c(
  prior(normal(5, 2), class = "Intercept"),
  prior(normal(0, 1), class = "b", coef = "age_z"),
  prior(normal(0, 1), class = "b", coef = "waist_z"),
  prior(normal(0, 1), class = "b"),   # covers race dummy coefficients
  prior(exponential(1), class = "sigma")
)

fit_nonhier <- brm(
  formula = nonhier_formula,
  prior = nonhier_priors,
  data = nhgh_model,
  chains = 4,
  iter = 2000,
  warmup=1000
)


summary(fit_nonhier)


nutsparams_fit_nonhier <- nuts_params(fit_nonhier)

sprintf("Divergence of the model: %i", sum(subset(nutsparams_fit_nonhier, Parameter == "divergent__")$Value))
```

Fitting and looking at convergence diagnostics for the hierarchical model with race variation:

```
hier_formula <- bf(
  gh ~ 1 + age_z + waist_z + (1 + age_z + waist_z | race),
  family = "student",
  center = FALSE
)

hier_priors <- c(
  prior(normal(5, 2), class = "b", coef = "Intercept"),
  prior(normal(0, 1), class = "b", coef = "age_z"),
  prior(normal(0, 1), class = "b", coef = "waist_z"),

  prior(exponential(1), class = "sd", group = "race", coef = "Intercept"),
  # weakly informative? also favors small SD values (gh -> does not have large variation e.g 5.5% -> 6.7%
  prior(exponential(1), class = "sd", group = "race", coef = "age_z"),
  prior(exponential(1), class = "sd", group = "race", coef = "waist_z"),
  prior(lkj(2), class = "cor", group = "race"),

  prior(exponential(1), class = "sigma") # weakly informative?
)

fit_hier <- brm(
  formula = hier_formula,
  prior = hier_priors,
  data = nhgh_model,
  chains = 4,
  iter = 2000,
  warmup=1000,
```

```
  control = list(adapt_delta = 0.99)
)


summary(fit_hier)


nutsparams_fit_hier <- nuts_params(fit_hier)

sprintf("Divergence of the model: %i", sum(subset(nutsparams_fit_hier, Parameter == "divergent__")$Value))


#conditional effect plot:
conditions_fit_hier <- make_conditions(nhgh_model, "race", incl_vars = FALSE)
me_fit_hier <- conditional_effects(fit_hier, conditions = conditions_fit_hier,
                          re_formula = NULL)
plot(me_fit_hier, ncol = 6, points = TRUE)
```

Fitting and looking at convergence diagnostics for the hierarchical model with both race and income variation:

```
hier_formula_inc_race <- bf(
  gh ~ 1 + age_z + waist_z +
    (1 + age_z + waist_z | race) +
    (1 | income),
  family = "student",
  center = FALSE
)

hier_priors_inc_race <- c(
  prior(normal(5, 2), class = "b", coef = "Intercept"),
  prior(normal(0, 1), class = "b", coef = "age_z"),
  prior(normal(0, 1), class = "b", coef = "waist_z"),

  prior(exponential(1), class = "sd", group = "race", coef = "Intercept"),
  prior(exponential(1), class = "sd", group = "race", coef = "age_z"),
  prior(exponential(1), class = "sd", group = "race", coef = "waist_z"),
  prior(lkj(2), class = "cor", group = "race"),

  prior(exponential(1), class = "sd", group = "income", coef = "Intercept"),
  prior(exponential(1), class = "sigma")
)

fit_hier_inc_race <- brm(
  formula = hier_formula_inc_race,
  prior = hier_priors_inc_race,
  data = nhgh_model,
  chains = 4,
  iter = 2000,
  warmup = 1000,
  control = list(adapt_delta = 0.99)
)


summary(fit_hier_inc_race)


nutsparams_fit_hier_inc_race <- nuts_params(fit_hier_inc_race)

sprintf("Divergence of the model: %i", sum(subset(nutsparams_fit_hier_inc_race, Parameter == "divergent__")$Value))
```

Fitting and looking at convergence diagnostics for the hierarchical model with both race and income variation:

```
hier_formula_inc <- bf(
  gh ~ 1 + age_z + waist_z + (1 | income),
  family = "student",
  center = FALSE
)
```

```
hier_priors_inc <- c(
  prior(normal(5, 2), class = "b", coef = "Intercept"),
  prior(normal(0, 1), class = "b", coef = "age_z"),
  prior(normal(0, 1), class = "b", coef = "waist_z"),

  prior(exponential(1), class = "sd", group = "income", coef = "Intercept"),
  prior(exponential(1), class = "sigma")
)

fit_hier_inc <- brm(
  formula = hier_formula_inc,
  prior = hier_priors_inc,
  data = nhgh_model,
  chains = 4,
  iter = 2000,
  warmup = 1000,
  control = list(adapt_delta = 0.99)
)


summary(fit_hier_inc)


nutsparams_fit_hier_inc <- nuts_params(fit_hier_inc)

sprintf("Divergence of the model: %i", sum(subset(nutsparams_fit_hier_inc, Parameter == "divergent__")$Value))
```

Posterior predictive checks:

```
#baseline
ppc0 <- pp_check(fit_intercept) + ggplot2::xlab("Baseline model")
+ ggplot2::coord_cartesian(xlim = c(2, 10))

#non-hierarchical
ppc1 <- pp_check(fit_nonhier)
+ ggplot2::xlab("Pooled model") + ggplot2::coord_cartesian(xlim = c(2, 10))
ppc11 <- pp_check(fit_nonhier, type="dens_overlay_grouped", group="race")
+ ggplot2::xlab("Pooled model divided into racial groups")
+ ggplot2::coord_cartesian(xlim = c(2, 10))

#hierarchical based on race
ppc2 <- pp_check(fit_hier, allow_new_levels=TRUE,sample_new_levels="gaussian") +
ggplot2::xlab("Hierarchical model based on race") +
ggplot2::coord_cartesian(xlim = c(2, 10))
ppc21 <- pp_check(fit_hier, type="dens_overlay_grouped", group="race",allow_new_levels=TRUE, sample_new_levels="gaussian")+
ggplot2::xlab("Hierarchical model based on race, divided into racial groups")
+ ggplot2::coord_cartesian(xlim = c(2, 10))

#hierarchical based on race & income
ppc3 <- pp_check(fit_hier_inc_race, allow_new_levels=TRUE,sample_new_levels="gaussian") +
ggplot2::xlab("Hierarchical model based on race & income") +
ggplot2::coord_cartesian(xlim = c(2, 10))

#hierarchical based on income
ppc4 <- pp_check(fit_hier_inc, allow_new_levels=TRUE,sample_new_levels="gaussian") +
ggplot2::xlab("Hierarchical model based on income") +
ggplot2::coord_cartesian(xlim = c(2, 10))
ppc41 <- pp_check(fit_hier_inc, type="dens_overlay_grouped", group="income",allow_new_levels=TRUE, sample_new_levels="gaussian")+
ggplot2::xlab("Hierarchical model based on income, divided into income groups")
+ ggplot2::coord_cartesian(xlim = c(2, 10))

ppc0

grid.arrange(ppc1, ppc2, ppc3, ppc4, ncol = 2)

ppc11

ppc21

ppc41
```

Model comparison (PSIS-LOO)

```
loo_intercept <- loo(fit_intercept)
loo_nonhier <- loo(fit_nonhier)
loo_hier_race <- loo(fit_hier)
loo_hier_inc <- loo(fit_hier_inc)
loo_hier_inc_race <- loo(fit_hier_inc_race)

pareto_k_table(loo_intercept)
pareto_k_table(loo_nonhier)
pareto_k_table(loo_hier_race)
pareto_k_table(loo_hier_inc)
pareto_k_table(loo_hier_inc_race)

# printing individual loo for each model
loo_intercept
loo_nonhier
loo_hier_race
loo_hier_inc
loo_hier_inc_race

# loo comparison between models
loo_compare(
  loo_intercept,
  loo_nonhier,
  loo_hier_race,
  loo_hier_inc,
  loo_hier_inc_race
)
```

Sensitivity analysis (priorsense)

```
# Key fixed effects we care about
vars_fixed_main <- c(
  "b_Intercept",
  "b_age_z",
  "b_waist_z"
)

# Key group-level SDs (race & inocme)
vars_group_main <- c(
  "sd_race__Intercept",
  "sd_race__age_z",
  "sd_race__waist_z",
  "sd_income__Intercept"
)

# distributional parameters
vars_dist_main <- c("sigma", "nu")

vars_main <- c(vars_fixed_main, vars_group_main, vars_dist_main)
vars_main

ps_data_main <- create_priorsense_data(fit_hier_inc_race)

ps_main <- powerscale_sensitivity(
  ps_data_main,
  variable  = vars_main,
  component = c("prior", "likelihood"),
  lower_alpha = 0.99,
  upper_alpha = 1.01,
  sensitivity_threshold = 0.05
)

ps_main

vars_fixed_race <- c(
  "b_Intercept",
  "b_age_z",
  "b_waist_z"
)

vars_group_race <- c(
```

```
  "sd_race__Intercept",
  "sd_race__age_z",
  "sd_race__waist_z"
)

vars_dist_race <- c("sigma", "nu")

vars_race <- c(vars_fixed_race, vars_group_race, vars_dist_race)
vars_race

ps_data_race <- create_priorsense_data(fit_hier)

ps_race <- powerscale_sensitivity(
  ps_data_race,
  variable  = vars_race,
  component = c("prior", "likelihood"),
  lower_alpha = 0.99,
  upper_alpha = 1.01,
  sensitivity_threshold = 0.05
)

ps_race

vars_fixed_nonhier <- c(
  "b_Intercept",
  "b_age_z",
  "b_waist_z"
)

vars_race_nonhier <- c(
  "b_raceOtherHispanic",
  "b_raceNonMHispanicWhite",
  "b_raceNonMHispanicBlack",
  "b_raceOtherRaceIncludingMultiMRacial"
)

vars_dist_nonhier <- c("sigma", "nu")

vars_nonhier <- c(vars_fixed_nonhier, vars_race_nonhier, vars_dist_nonhier)
vars_nonhier

ps_data_nonhier <- create_priorsense_data(fit_nonhier)

ps_nonhier <- powerscale_sensitivity(
  ps_data_nonhier,
  variable  = vars_nonhier,
  component = c("prior", "likelihood"),
  lower_alpha = 0.99,
  upper_alpha = 1.01,
  sensitivity_threshold = 0.05
)

ps_nonhier

vars_fixed_inc <- c(
  "b_Intercept",
  "b_age_z",
  "b_waist_z"
)

vars_group_inc <- c(
  "sd_income__Intercept"
)

vars_dist_inc <- c("sigma", "nu")

vars_inc <- c(vars_fixed_inc, vars_group_inc, vars_dist_inc)
vars_inc

ps_data_inc <- create_priorsense_data(fit_hier_inc)
```

```
ps_inc <- powerscale_sensitivity(
  ps_data_inc,
  variable  = vars_inc,
  component = c("prior", "likelihood"),
  lower_alpha = 0.99,
  upper_alpha = 1.01,
  sensitivity_threshold = 0.05
)

ps_inc

vars_intercept <- c(
  "b_Intercept",
  "sigma",
  "nu"
)

ps_data_intercept <- create_priorsense_data(fit_intercept)

ps_intercept <- powerscale_sensitivity(
  ps_data_intercept,
  variable  = vars_intercept,
  component = c("prior", "likelihood"),
  lower_alpha = 0.99,
  upper_alpha = 1.01,
  sensitivity_threshold = 0.05
)

ps_intercept
```

Plots to visualise the sensitivity analysis

```
#prior power-scaling: densities for age and waist effects (fit_hier_inc_race)
ps_seq_prior_main <- powerscale_sequence(
  ps_data_main,
  component = "prior"
)

powerscale_plot_dens(
  ps_seq_prior_main,
  variable = c("b_age_z", "b_waist_z")
)

#prior power scaling: mean/SDs for race and income SDs
powerscale_plot_quantities(
  ps_seq_prior_main,
  variable  = c("sd_race__Intercept", "sd_income__Intercept"),
  quantity  = c("mean", "sd"),
  div_measure = "cjs_dist"
)

#likelihood power scaling: age effect
ps_seq_lik_main <- powerscale_sequence(
  ps_data_main,
  component = "likelihood"
)

powerscale_plot_quantities(
  ps_seq_lik_main,
  variable  = "b_age_z",
  quantity  = c("mean", "sd"),
  div_measure = "cjs_dist"
)

#hier_race model:
ps_seq_prior_race <- powerscale_sequence(
  ps_data_race,
  component = "prior"
)

powerscale_plot_dens(
```

```
  ps_seq_prior_race,
  variable = c("b_age_z", "b_waist_z")
)

#nonhier model:
ps_seq_prior_nonhier <- powerscale_sequence(
  ps_data_nonhier,
  component = "prior"
)

powerscale_plot_dens(
  ps_seq_prior_nonhier,
  variable = c("b_age_z", "b_waist_z")
)

powerscale_plot_quantities(
  ps_seq_prior_nonhier,
  variable  = vars_race_nonhier,
  quantity  = "mean"
)

#hier_inc model:
ps_seq_prior_inc <- powerscale_sequence(
  ps_data_inc,
  component = "prior"
)

powerscale_plot_dens(
  ps_seq_prior_inc,
  variable = c("b_age_z", "b_waist_z")
)

powerscale_plot_quantities(
  ps_seq_prior_inc,
  variable  = "sd_income__Intercept",
  quantity  = c("mean", "sd")
)

#Baseline intercept model:
ps_seq_prior_intercept <- powerscale_sequence(
  ps_data_intercept,
  component = "prior"
)

powerscale_plot_dens(
  ps_seq_prior_intercept,
  variable = c("b_Intercept")
)

powerscale_plot_quantities(
  ps_seq_prior_intercept,
  variable = c("sigma", "nu"),
  quantity = c("mean", "sd")
)
```